VIV INTERCHEM

VIVAFLOC CP2920

Revision: 01

Revision date 29/11/2024

SECTION 1: Identification of the substance/mixture and of the company/undertaking		
1.1	**Product identifier**	VIVAFLOC CP2920
1.2	**Relevant identified uses of the substance or mixture and uses advised against**	
	Relevant identified uses:	Flocculation agent
1.3	**Details of the supplier of the safety data sheet**	
	Company:	VIV INTERCHEM 22 Sukhumvit 42, Sukhumvit Rd, Prakanong, Klongtoey, Bangkok 10110 Telephone: +662 712 1044 E-mail address: thasanaprapha@vivgroup.com
1.4	**Emergency telephone number**	
	International emergency number:	Telephone: 02-7094600-4 #214

SECTION 2: Hazards Identification	
2.1	**Classification of the substance or mixture**
	According to Regulation (EC) No 1272/2008 [CLP] No need for classification according to GHS criteria for this product.
2.2	**Label elements**
	According to Regulation (EC) No 1272/2008 [CLP] The product does not require a hazard warning label in accordance with GHS criteria.
2.3	**Other hazards**
	According to Regulation (EC) No 1272/2008 [CLP] Very slippery when wet. This type of product has a tendency to create dust if roughly handled. The product does not burn readily but as with many organic powders, flammable dust clouds may be formed in air. The product is under certain conditions capable of dust explosion.

SECTION 3: Composition/information on ingredients		
3.1	**Substances**	Not applicable
3.2	**Mixtures**	Polyacrylamide, cationic

SECTION 4: First aid measures		
4.1	**Description of first aid measures**	
	If inhaled:	If difficulties occur after dust has been inhaled, remove to fresh air and seek medical attention.

	On skin contact:	Wash thoroughly with soap and water.
	On contact with eyes:	Wash affected eyes for at least 15 minutes under running water with eyelids held open.
	On ingestion:	Rinse mouth and then drink plenty of water. Check breathing and pulse. Place victim in the recovery position, cover and keep warm. Loosen tight clothing such as a collar, tie, belt or waistband. Seek medical attention. Never induce vomiting or give anything by mouth if the victim is unconscious or having convulsions.
4.2	**Most important symptoms and effects, both acute and delayed**	
	Symptoms:	No significant symptoms are expected due to the non-classification of the product.
	Hazards:	No hazard is expected under intended use and appropriate handling.
4.3	**Indication of any immediate medical attention and special treatment needed**	
	Treatment:	Treat according to symptoms (decontamination, vital functions), no known specific antidote.

SECTION 5: Firefighting measures

5.1	**Extinguishing media**	
	Suitable extinguishing media:	dry powder, foam
	Unsuitable extinguishing media for safety reasons:	water jet, carbon dioxide
	Additional information:	If water is used, restrict pedestrian and vehicular traffic in areas where slip hazard may exist.
5.2	**Special hazards arising from the substance or mixture**	
	carbon oxides, nitrogen oxides.	The substances/groups of substances mentioned can be released in case of fire. Very slippery when wet.
5.3	**Advice for fire-fighters**	
	Special protective equipment:	Wear a self-contained breathing apparatus.
	Further information:	The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

SECTION 6: Accidental release measures

	Avoid dispersal of dust in the air (i.e., clearing dust surfaces with compressed air). Avoid the formation and build-up of dust - danger of dust explosion. Dust in sufficient concentration can result in an explosive mixture

	in air. Handle to minimize dusting and eliminate open flame and other sources of ignition. Forms slippery surfaces with water.	
6.1	**Personal precautions, protective equipment and emergency procedures**	Use personal protective clothing.
6.2	**Environmental precautions**	Do not discharge into drains/surface waters/groundwater.
6.3	**Methods and material for containment and cleaning up**	
	For small amounts:	Pick up with suitable appliance and dispose of.
	For large amounts:	Contain with dust binding material and dispose of. Avoid raising dust.
6.4	**Reference to other sections**	
	Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.	

SECTION 7: Handling and storage

7.1	**Precautions for safe handling**	Breathing must be protected when large quantities are decanted without local exhaust ventilation. Handle in accordance with good industrial hygiene and safety practice. Forms slippery surfaces with water.
	Protection against fire and explosion:	Avoid dust formation. Dust in sufficient concentration can result in an explosive mixture in air. Handle to minimize dusting and eliminate open flame and other sources of ignition. Dry powders can build static electricity charges when subjected to the friction of transfer and mixing operations. Provide adequate precautions, such as electrical grounding and bonding, or inert atmospheres.
7.2	**Conditions for safe storage, including any incompatibilities**	
	Further information on storage conditions:	Store in unopened original containers in a cool and dry place. Avoid wet, damp or humid conditions, temperature extremes and ignition sources.
	Storage stability:	Avoid extreme heat.
7.3	**Specific end use(s)**	For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

SECTION 8: Exposure controls/personal protection

8.1	**Components with occupational exposure limits:**	
	Control parameters	Components with occupational exposure limits Particles, not otherwise specified, respirable Particles, not otherwise specified, inhalable
8.2	**Personal Protective Equipment：**	
	Eye/Face Protection:	Safety glasses with side-shields (frame goggles) (e.g. EN 166)

	Body Protection:	light protective clothing
	Hand protection:	Chemical resistant protective gloves (EN 374) Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374): e.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), polyvinylchloride (0.7 mm) and other Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation
	Respiratory Protection:	Suitable respiratory protection for lower concentrations or short-term effect: Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)
	General safety and hygiene measures:	Handle in accordance with good industrial hygiene and safety practice. Ensure adequate ventilation. Wearing of closed work clothing is recommended. No eating, drinking, smoking or tobacco use at the place of work.
	Environmental exposure controls:	For information regarding environmental exposure controls, see Section 6.

SECTION 9: Physical and chemical properties		
9.1	**Information on basic physical and chemical properties**	
	Appearance:	Powder
	Colour:	Off-White
	Odour:	Odourless
	Odour threshold:	No applicable information available.
	pH value:	2.5 – 4.5 (10 g/l)
	Melting range:	The substance / product decomposes therefore not determined.
	Boiling point:	Not applicable
	Flash point:	Not applicable
	Evaporation rate:	The product is a non-volatile solid.
	Flammability (solid/gas):	Not flammable
	Lower explosion limit:	For solids not relevant for classification and labelling.
	Upper explosion limit:	For solids not relevant for classification and labelling.
	Thermal decomposition:	No decomposition if stored and handled as prescribed/indicated.
	Vapour pressure:	The product has not been tested
	Solubility in water:	Forms a viscous solution.

	Partitioning coefficient n-octanol/water (log Kow):	Study scientifically not justified.
	Self ignition:	Not self-igniting
	Viscosity, dynamic:	Not determined
	Explosion hazard:	Not explosive
	Fire promoting properties:	Not fire-propagating
9.2	**Other information**	
	Self heating ability:	It is not a substance capable of spontaneous heating.
	Minimum ignition energy:	> 1 J
	Bulk density:	Approx. 800 kg/m3
	Hygroscopy:	Non-hygroscopic
	Other Information:	If necessary, information on other physical and chemical parameters is indicated in this section.

SECTION 10: Stability and reactivity		
10.1	**Conditions to avoid:**	Avoid extreme temperatures. Avoid humidity.
10.2	**Thermal decomposition:**	No hazardous decomposition products if stored and handled as prescribed/indicated.
10.3	**Substances to avoid:**	strong acids, strong bases, strong oxidizing agents
10.4	**Corrosion to metals:**	No corrosive effect on metal.
10.5	**Hazardous reactions:**	The product is not a dust explosion risk as supplied; however the build-up of fine dust can lead to a risk of dust explosions.
10.6	**Hazardous decomposition products:**	No hazardous reactions if stored and handled as prescribed/indicated.
10.7	**Chemical stability**	The product is stable if stored and handled as prescribed/indicated.
	Peroxides:	0 %, The product does not contain peroxides.

SECTION 11: Toxicological information		
11.1	**Acute toxicity**	
	Experimental/calculated data:	LD50 rat (oral): > 5,000 mg/kg (OECD Guideline 401)
11.2	**Irritation effects**	
	Experimental/calculated data:	Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404) Serious eye damage/irritation rabbit: non-irritant
11.3	**Respiratory/Skin sensitization**	

	Experimental/calculated data:	Based on the ingredients, there is no suspicion of a skin-sensitizing potential.
11.3	**Germ cell mutagenicity**	Based on the ingredients, there is no suspicion of a mutagenic effect.
11.4	**Carcinogenicity**	The whole of the information assessable provides no indication of a carcinogenic effect.
11.5	**Reproductive toxicity**	Based on the ingredients, there is no suspicion of a toxic effect on reproduction.
11.6	**Specific target organ toxicity (single exposure)**	No data available.
11.7	**Repeated dose toxicity and Specific target organ toxicity (repeated exposure)**	Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses. The product has not been tested. The statement has been derived from the properties of the individual components.
11.8	**Aspiration hazard**	No aspiration hazard expected.
11.9	**Other relevant toxicity information**	The product has not been tested. The statements on toxicology have been derived from products of a similar structure and composition.

SECTION 12: Ecological information

12.1	**Toxicity**	
	Assessment of aquatic toxicity:	Fish toxicity and aquatic toxicity are drastically reduced by rapid irreversible adsorption onto suspended and/or dissolved organic matter. Acute effects on aquatic organisms are due to the cationic charge of the polymer, which is quickly neutralized in natural water courses by irreversible adsorption onto particles, hydrolysis and dissolved organic carbon. The hydrolysis products are not acutely harmful to aquatic organisms.
	Toxicity to fish:	LC50 (96 h) 10 - 100 mg/l, Fish (static)
	Aquatic invertebrates:	EC50 (48 h) 10 - 100 mg/l, daphnia
12.2	**Mobility**	
	Information on:	Cationic polyacrylamide
	Assessment transport between environmental compartments:	Adsorption in soil: Adsorption to solid soil phase is expected.
12.3	**Persistence and degradability**	
	Assessment biodegradation and elimination (H2O):	Not readily biodegradable (by OECD criteria).
	Information on Stability in Water (Hydrolysis):	> 70 % (28 d) (pH value > 6) In contact with water the substance will hydrolyse rapidly.

12.4	**Bioaccumulation potential**	
	Assessment bioaccumulation potential:	Based on its structural properties, the polymer is not biologically available. Accumulation in organisms is not to be expected.
12.5	**Results of PBT and vPvB assessment**	According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.
12.6	**Other adverse effects**	The product does not contain substances that are listed in Annex I of Regulation (EC) 2037/2000 on substances that deplete the ozone layer.
12.7	**Additional information**	
	Other ecotoxicological advice:	Must not be discharged into the environment. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

SECTION 13: Disposal considerations		
13.1	**Waste treatment methods**	Dispose of in accordance with national, state and local regulations.
	Contaminated packaging:	Packs that cannot be cleaned should be disposed of in the same manner as the contents. Uncontaminated packaging can be re-used.

SECTION 14: Transport information		
	Land transport	
	ADR, RID	Not classified as a dangerous good under transport regulations
	UN number:	Not applicable
	UN proper shipping name:	Not applicable
	Transport hazard class(es):	Not applicable
	Packing group:	Not applicable
	Environmental hazards:	Not applicable
	Special precautions for user	None known
	Inland waterway transport	
	ADN	Not classified as a dangerous good under transport regulations
	UN number:	Not applicable

	UN proper shipping name:	Not applicable
	Transport hazard class(es):	Not applicable
	Packing group:	Not applicable
	Environmental hazards:	Not applicable
	Special precautions for user	None known
	Transport in inland waterway vessel	Not evaluated
	Sea transport	
	IMDG	Not classified as a dangerous good under transport regulations
	UN number:	Not applicable
	UN proper shipping name:	Not applicable
	Transport hazard class(es):	Not applicable
	Packing group:	Not applicable
	Environmental hazards:	Not applicable
	Special precautions for user	None known
	Air transport	
	IATA/ICAO	Not classified as a dangerous good under transport regulations
	UN number:	Not applicable
	UN proper shipping name:	Not applicable
	Transport hazard class(es):	Not applicable
	Packing group:	Not applicable
	Environmental hazards:	Not applicable
	Special precautions for user	None known
14.1	**UN number**	See corresponding entries for “UN number” for the respective regulations in the tables above.
14.2	**UN proper shipping name**	See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3	**Transport hazard class(es)**	See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.
14.4	**Packing group**	See corresponding entries for “Packing group” for the respective regulations in the tables above.
14.5	**Environmental hazards**	See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.
14.6	**Special precautions for use**	See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.
14.7	**Transport in bulk according to Annex II of MARPOL and the IBC Code**	
	Regulation:	Not evaluated
	Shipment approved:	Not evaluated
	Pollution name:	Not evaluated
	Pollution category:	Not evaluated
	Ship Type:	Not evaluated

SECTION 15: Regulatory information	
15.1	**Safety, health and environmental regulations/legislation specific for the substance or mixture**
15.2	**Chemical Safety Assessment**
	Chemical Safety Assessment not yet performed due to registration timelines

SECTION 16: Other information		
	Assessment of the hazard classes according to UN GHS criteria (most recent version)	
	Aquatic Acute 3 Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:	
	Eye Dam./ Irrit.	Serious eye damage/eye irritation
	H319	Causes serious eye irritation.
	H318	Causes serious eye damage.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. The data do not describe the product's properties (product specification). Neither should any agreed property nor the suitability of the product for any specific purpose be deduced from the data contained in the safety data sheet. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

END OF DATA SHEET.